

*This thesis proposes a territorial decentralisation strategy
for the Ohrid UNESCO region, redistributing urban
development pressure from the shoreline to the region's rural
settlements through transformation, renewal and
adaptive-reuse, as an alternative to the market-led urban
expansion currently threatening its World Heritage status*

TERRITORIAL TRANSFORMATIONS

The Ohrid region is a UNESCO World Heritage site shaped by millennia of nature and human culture. Tourism has become the area's primary industry, attracting half a million visitors annually, a number that continues to grow rapidly. While this brings economic benefits, it also places significant pressure on the region's heritage, producing poorly planned urban development and a market-led architectural language without meaningful heritage integration. As new building concentrates in the city and along the shoreline, the existing stock in the region's smaller settlements sits empty and non-valorised, even where it carries genuine heritage value.

Given this territorial complexity, the thesis adopts a mixed methodology combining qualitative analysis with an iterative, multiscale design approach.

In response, it proposes a transformation strategy built on decentralisation rather than centralised urban growth, and on the thoughtful renewal of existing structures. By redistributing development across the region's diverse settlements and reusing their building stock while preserving each one's spatial character, the strategy aims to relieve pressure on the protected landscape while actively reinforcing the values for which the region was inscribed.







Personal Motivation

The interest in this topic traces back to 2019, in my final year of high school in my hometown of Ohrid in Macedonia, where I developed a particular affinity for the town's overlooked regional modernist gems.. There was a special interest towards overlooked regional modernists gems in the town. Two years after that, I had started my studies in architecture and, one of those sites, positioned sensitively on the shoreline was torned down, so instead a real-estate property could be developed. This led to a rather dissapointing feeling, and I was confused on how could this happen? How was the value of the building not considered? From that point on, it was an ever expansive curiosity...

As it turn out, we shared a similar feeling with some other colleagues, which led to a founding of an architectural collective, still active, scanning, documenting and promoting the less-loved but full of potential "future heritage" sites and places, compared to the overly exploited "UNESCO labeled" landmarks.

Pursuing my curiosity I landed in Denmark, for the Strategic Design & Entrepreneurship master studies, looking for answers in the architecture and planning sphere. Here, as well as home, there are number of challenges, regarding the built industry, but on the other hand, there is also quite an extensive knowledge and care for it. In Copenhagen, I am introduced to new narratives, technologies, methodologies and means that are shaping the global architectural and planning discourse. As an active observant and student I feel that the best way to integrate them in my knowledge base is to test them out and explore them in a context that I am most familiar with-that being the Ohrid UNESCO WH region.

Global Relevance

Cities are growing, urban zones are taking over, producing distance between humans and nature. By 2050 68% percent of the population is expected to live in urban zones (United Nations, 2018).

Looking at the 2030 UN SDGs agenda, this thesis orients itself to produce knowledge and support first and foremost Goal 11, targeting heritage protection and more sustainable alternatives for the current urban development state, Goal 09, promoting a infrastructure and settlement network with more sustainable development potentials, and Goal 17- to strengthen the means of implementation and revitalize the local partnerships, but also potential for cross-boundary development, with the selected region lying on the border, and being part of the larger Ohrid-Prespa UNESCO Transboundary Biosphere Reserve.

Professional Challenges

As the world is experiencing shifts, so it is architecture. Automatization, machine learning models, economic systems and market forces are some of the things that shape the profession today. Through this thesis, as well as with the Strategic Design and Entrepreneurship programme we question where the role of the architect can be placed. Which fields can architecture support and use it's means of production, such as conceptualizing, ideation, visual communication, etc. In which fields can architectural knowledge migrate and be useful?



CONTEXT

where?



As a case study the Ohrid region was selected. It is part of the greater Ohrid-Prespa UNESCO Transboundary Biosphere reserve, as well as The Ohrid UNESCO culture and nature region in Southeast Europe, that are positioned within the Adriatic Basin.

Understanding the importance of the site we find ourselves at the lakes formation, between 1.9 and 1.3 million years ago, from when it starts its continuum, experiencing significant environmental changes, but somehow showing high resilience of it's ecosystem, with relatively low extinction rates, explaining it's rich biodiversity and high value (Wagner et al., 2017, p. 2049), today being a home to over 200 endemic species (Albrecht & Wilke, 2008; UNESCO, 2024).

Lake Ohrid, with strong strategic positioning and resources is the oldest (Wagner et al., 2014, p. 2), and most heritage (both cultural and environmental) sensitive lake in Europe, functions as both a hydrological system and a cultural archive, supporting endemic biodiversity and continuous human settlement for over 8,000 years (Yermokhin et al., 2025, p. 11) labelled with UNESCO OUV criteria (i), (ii), (iii) and (vi), making it a World Heritage-WH site.

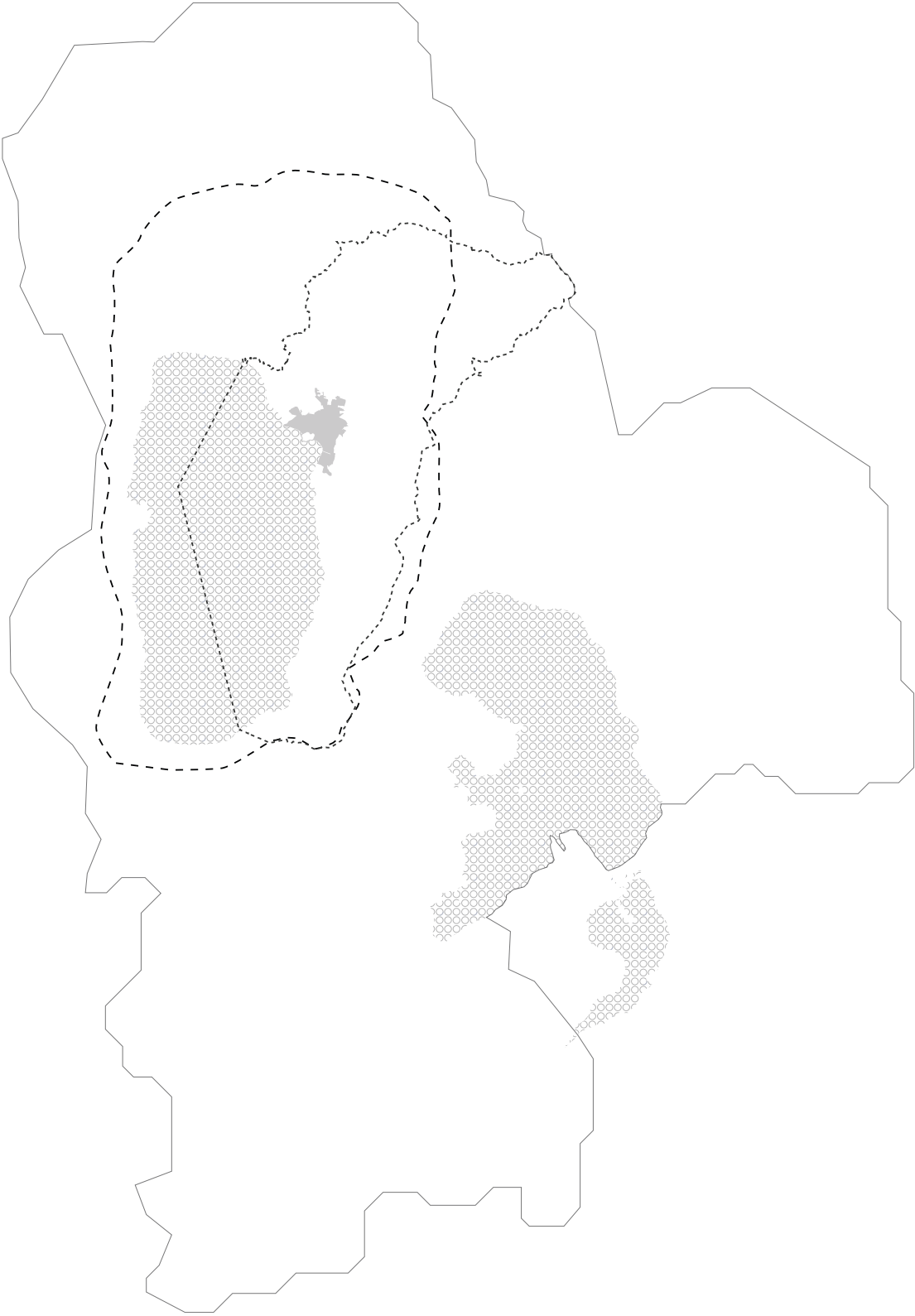
Over time and shaped by sociopolitical factors, the region has steadily gained national and international acknowledgment for tourism, leading for it to be one of the dominant industries.



Case location on a map of Europe, by author, 2026



Case location showing proximity between Macedonia, Albania and Greece, by author, 2026



different Ohrids?

- Municipality of Ohrid
- Lakes
- Ohrid old town core
- Greater Ohrid
- Ohrid UNESCO Natural and Cultural WH region
- Ohrid Prespa UNESCO Transboundary Biosphere

10km



photos depicting the regions atmosphere, by author, 2023



photos depicting the regions atmosphere, by author, 2025



photos depicting the regions atmosphere, by author, 2026



photos depicting the regions atmosphere, by author, 2025



photos depicting the regions atmosphere, by author, 2026



photos depicting the regions atmosphere, by author, 2025



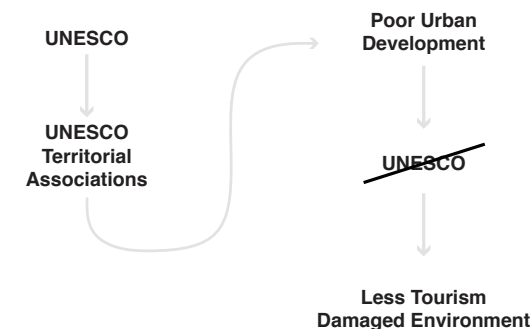
CHALLENGE

what?



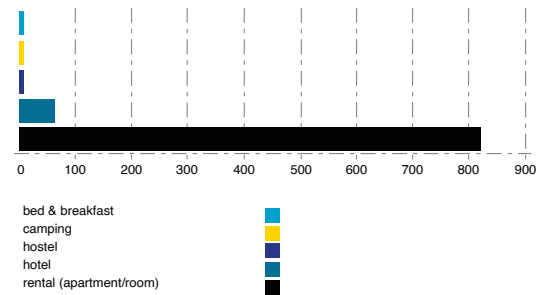
Tourism, the territorial associations (Kärrholm, 2007) with UNESCO, together with its attractive position for urban development investment, as well as with human behavior and market forces, leads to pressures on the environment. As informed from the feasibility report of the shoreline (Thomson, D'Andrea, 2024), the apartment housing typologies are oversaturated and ever growing. On another note on the map we are able to see how much the city is being more urbanized, more built, more developed with the same typologies, and generic market-led materials, while already built building stock in the backbone of the region, the rural areas, hinterlands and smaller settlements sit empty and non-valorized. Some of these houses are built in a traditional and vernacular manner, making them part of UNESCO OUV criterion (i)

"The concentration of the archaeological remains and urban structures within the old urban centre of Ohrid, in the Lin Peninsula, and along the coast of Lake Ohrid as well as the surrounding areas creates an exceptional harmonious ensemble, which is one of the key features that make this region truly unique" (UNESCO WHC, n.d.)



UNESCO associations diagrams, by autor, 2026

Contributing to the conclusion that **poorly planned urban development and generic, market-led architectural language is a heavy pressure on the heritage landscape**, leading to warning and potential listing on UNESCO World Heritage sites in danger (UNESCO, ICOMOS, IUCN, 2019). As new developments tend to be developed with maximum square meters at the lowest price, putting the heritage, such as the organically developed urban morphology, original programs, local architectural elements, traditional building technology and know-how as well as local materials aside and as a secondary, tertiary thing or sometimes banalizing it into mimicked decorative elements.



visitor's accommodation mapping and typology, data collected by the FS2024 research team in April 2023 and analysed for the study

This has been also supported during the fieldtrip and interviews conducted by the local actors, which will be better elaborated later. This opens up certain questions that we might explore to generate knowledge and better understand the problem, such as:

How can architecture and planning strategies impact for a more sustainable and sensitive development in a coastal heritage landscape?

Can decentralization and better connection between settlements better support their heritage values?

How can adaptive reuse strategies transform vernacular existing buildings into spatial and economic resources, enabling affordable growth without degrading the water-dependent cultural and environmental heritage that attracts tourism?

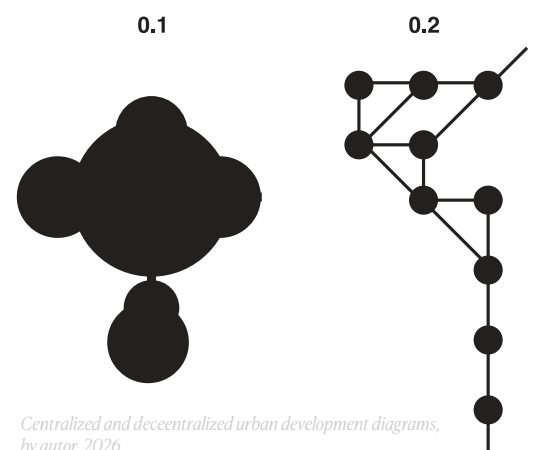
What could reuse of building stock mean in a broader territorial context? Can reuse replace expansion as the driver of growth in a water-heritage landscape?

Scenario 0.1

business as usual; even bigger decentralisation; poor urban development "eats" unique settlements; architectural integrity slowly disappears, together with that the uniqueness and its UNESCO status...

Scenario 0.2

decentralisation strategy; more distributed network; public transport connections; reuse and integration of unique architectural language and keeping alive local building knowhow...



Centralized and decentralized urban development diagrams, by autor, 2026



STAKEHOLDERS

who?



What about the locals, what are their thoughts....?

During the fieldtrips several interviews were conducted with several actants in the region, mostly professionals. Through semi-structured and structured onsite interviews, data was gathered on past, current and future oriented challenges, actions and overall thoughts about the regions past, current and future development initiatives. Through Qualitative Analysis, which consisted of coding, data visualization and categorization, six major challenges were identified, *corruption, urbanisation, isolation (of the region), mismanagment, environmenal degradation and depopulation*, prioritizing to work with urbanisation, where the disciplinary tools of architecture and design are most applicable. During the interviews it was confirmed that the challenge that has been hypostatized was not far from the current positioning.

As described by the interviewees:

...the main challenge is the lack of human resource, technological equipment and knowledge...

...there aren't enough local craftsman that use the traditional know-how...

...often the new "traditionally" looking houses are build with concrete, steel and bricks, which are available on the market...

-Heritage Management expert

... This region is usually a transitional route, where people come for one day, they visit the Ohrid old town, then they go to St. Naum springs in the second day, and on the third day they escape from us away to Skopje or Bitola, we need to extend their stay! ...

-Toursim expert

...architecture and planning are seen as an investment tool, and nothing beyond...

...we cannot build mimicking the traditional vernacular architecture in a new context, you must be liberated and build in a more contemporary context...

...conservation sector is limiting the creative freedom of the planners and architects...

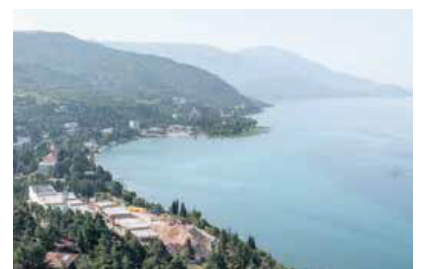
-Urban Planning expert

On the question what they see as a good and bad example, all of them suggested that the old town of Ohrid is the perfect example, where the planner, while confirming the organic urban morphology as a nice text, he posits that the "words" are not properly written, emphasizing the material choices and building techniques.

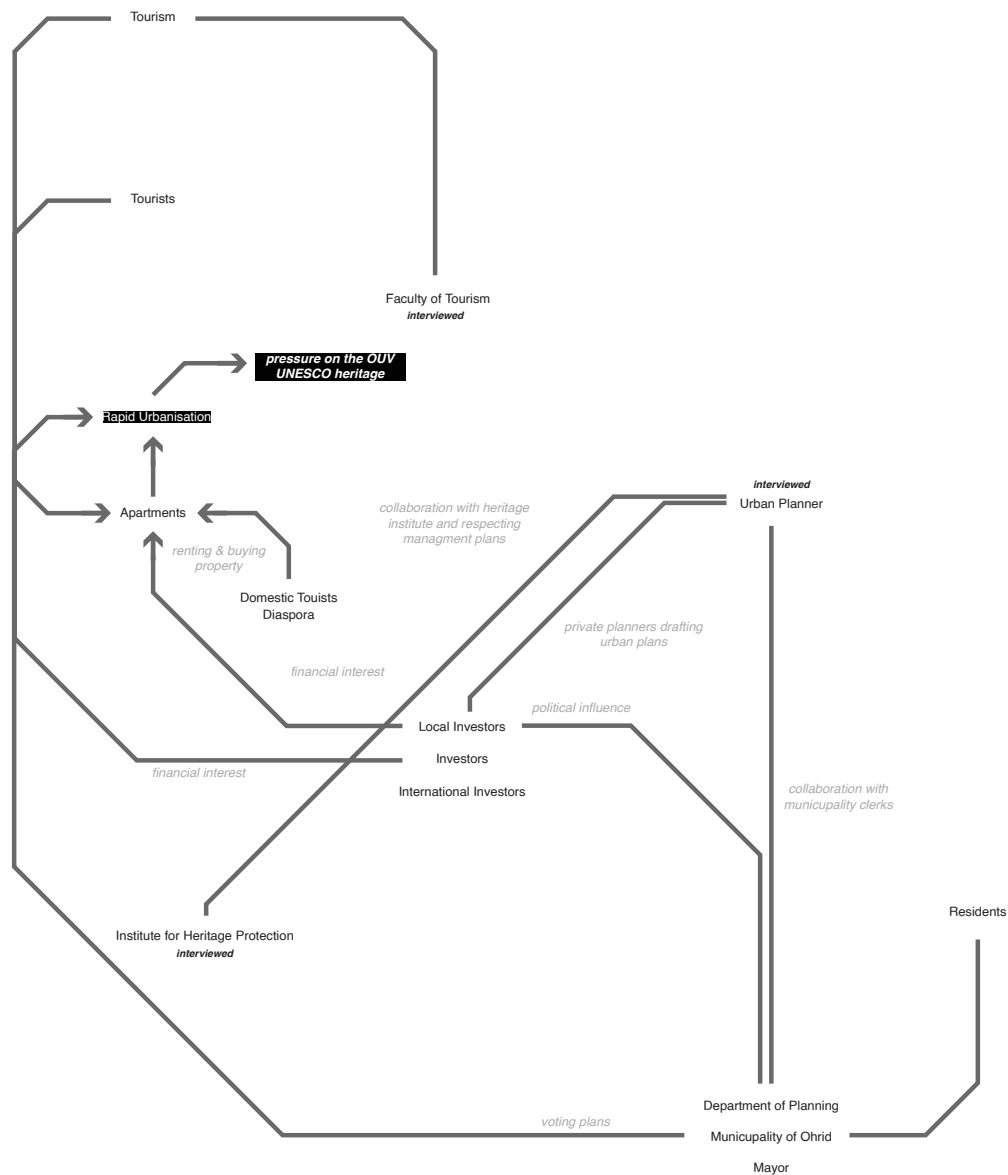
As a bad example, nearly all of them complained regarding the new apartment housing typologies as an alien and market led architecture, that has no place in the environment, where the tourism expert further elaborated that the same typologies are also reducing tourism sector, as people that would have gone to the hotels are simply just buying them and renting them. Here, the heritage management expert pointed out to gentrification as a challenge alongside.



what has been described as a positive example of urban fragment by interviewees, by author 2023



what has been described as a negative example by interviewees, by author 2025



During the site visits and interviews this associative map was reconstructed showing the intertwined powerdynamics and points of interests.

This can help us trace the planning and building decision making process in this region.



How can this challenge be approached?

Methodology

Given the contextual complexity, this research adopted a qualitative, mixed methods methodology, grounded in iterative design and multiscalar approach. It looks into strategic planning on a specific territory as something to be looked at, and analysed through different perspectives, as it is harder to quantify and estimate it through a single discipline, which might be sometimes constraining.

Methods

For data gathering and analysis the following techniques were deployed: literature review, policy and report review, semi-structured and structured interviews, fieldwork, and documentation, such as photography and video, which complemented the limited data given the regions peripheral positioning, qualitative analysis, cartographic and typological mapping, case study architecture approach.

Process

0.1 Choosing the context with the triple problem-based tool;

As explained in the introduction, the choice of topic and location was supported with the triple-based problem tool (Møller, 2025), looking at global and local challenges, while also exploring the professional discourse-architecture.

0.2 Problem statement and research question.

With background data gathering and research production from SDE3 course, looking into a broader Ohrid-Prespa UNESCO Bioregion reserve, it was narrowed down to Ohrid UNESCO region, due to its challenge with urbanisation and tourism, which is more prominent there, more specifically it's settlements, proximity to the lake and their urban development.

How can water-integrated architecture and planning territorial strategies transform and renew heritage-sensitive landscapes?

0.3 Project planning and programme;

The programme was developed and a timeline as a guiding document for the research.

0.4 On-site field trip and interview conducting;

During the past year three fieldtrips were made from Copenhagen to the Ohrid Region, two in the past semester, while one in this one, where 11 mixed online and on-site interviews were conducted, three on-site in this semester, all with involved professionals in the management and heritage related sectors from the region. During the fieldtrips, 14 settlements were visited (5 this semester, and 9 in the last) and 73 sites were mapped out in the Ohrid UNESCO WH region. With mixed documentation methods, such as GIS mapping analysis, drone footage, where a photographic and videographic archive was made, being generally very useful as being a provincial region there are not a lot of digitalized materials available.

0.5 Qualitative analysis (coding, sorting, comparing);

Later, the gathered dataset was carefully coded, while reading through it multiple times, after which the insights were thematically categorized into different themes to inform the challenge and problem statement. As a consequence of the fieldtrip notes and visits the settlements were analyzed, looking into their spatial and unique characteristics and their proximity to the lake, as well as their main economic models, and placed in 10 different typologies.

0.6 Iterative design process (research by design);

Iteratively throughout the whole process artistic methods and mediums were used to materialize the thinking and creative process, resulting in a collection of sketches, freehand & technical drawings, diagrams, virtual and physical models, in varying techniques of expression. This step was crucial for decision making and differentiating perspectives, as well as a productive tools for discussion and better understanding.

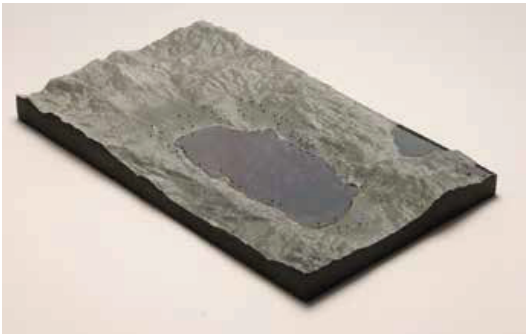
0.7 proposal

The iterative design process contributed with a proposal in three different scale, a territorial strategy- looking at the overall UNESCO WH region, an urban planning strategy- focusing on the fragment with the most threatening urban development pressure, as well as an exploration in an architectural scale, of one settlement as a case study. This workflow enabled to work between scales where one scale would test the decisions in the previous one. The latest proposal consists of a strategic model, communicated through diagrams, urban plan as well as scaled down architectural means and possible scenarios. (Appendix 3, proposal)

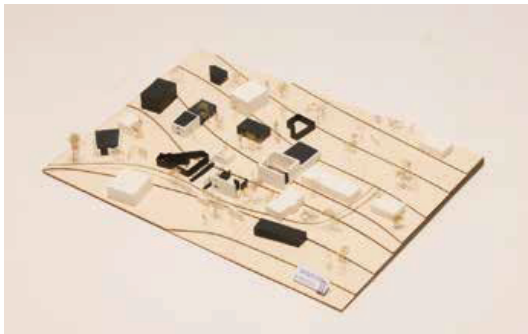
Through multiscale and open approach, testing different spatial configurations, their analysis and discussion, the decision process was informed and developed gradually, evolving to the current proposal, embodied and presented through maps, models, technical drawings and visualisations.



different architectural scenario iterations, by author 2026



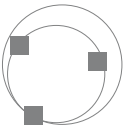
UNESCO WH region with it's settlements, by author 2026



latest model iteration, by author 2026

PROPOSAL

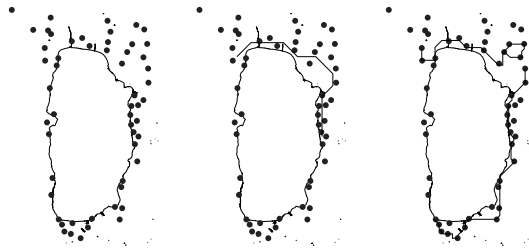
how?



While new rental apartments getting build more and more in the urban zone...already built stock in non-urban areas sleeps and waits to be woken up...

Strategy

The strategy proposes an alternative to the current developmental model where financial sustainability is prioritized, and looks forward into integrating more social and environmental sustainability as well (Doughnut for urban development, 2023). Could the market demand for housing be covered with reused infrastructure that would support the heritage values and ambient in the region, while also providing necessary structures and infrastructures to accommodate the industries? The proposal tries to identify the settlements' drawbacks and qualities, and to utilize their potential, as well as their integration with each other.



Speculative territorial strategy scenarios diagrams, by autor, 2026

Urban planning strategy

As a case study for the urban plan the northeast segment of the region was selected, as it is the one that has the most urbanization pressure and generic apartment housing typologies congestion (Thomson, D'Andrea, 2024). In this case the questions that are opened up are why build new and more apartment housing along the shoreline and centralizing, while built structures, integrated in the ambient with traditional building techniques are found in close proximity.

The plan proposes zoning scenarios, possible balanced urban and architecture programmes, while also looking at how each different types of settlements might work together, part of a larger decentralized, distributed network (Doughnut for Urban development, 2023), and would that be a more sustainable model to compete with the current centralized.



Speculative urban planning strategy scenarios diagrams, by autor, 2026

During the fieldtrips 73 settlements were identified, all of them within the UNESCO boundaries, each with its unique story, unfortunately overshadowed by the main town of Ohrid. This leads towards centralisation and congested pressure. But what if a decentralisation strategy was implemented?

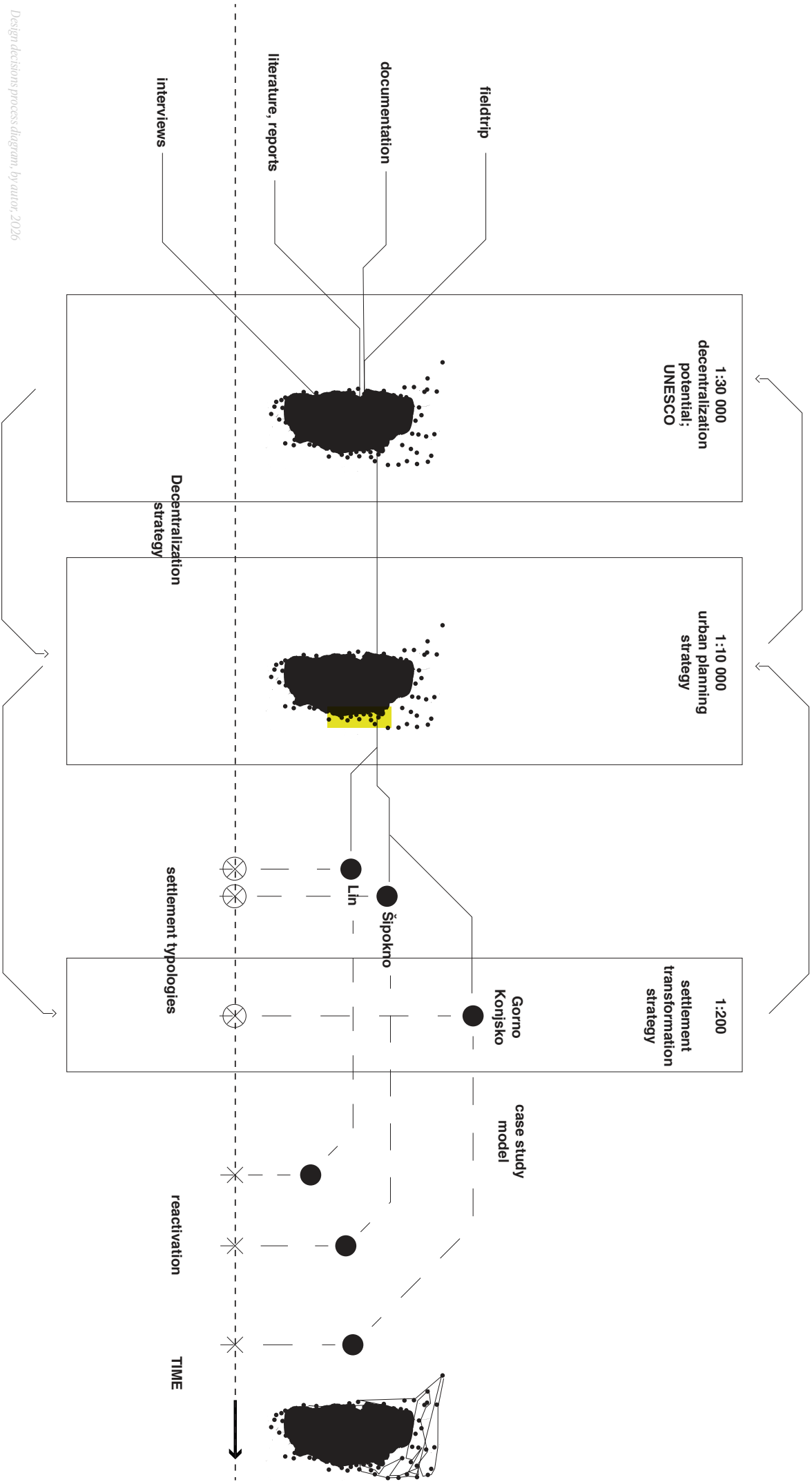
The zoning and scenario based proposal opens the discussion on how to develop in a heritage sensitive areas? Could the love of the locals and the market forces work together? Can the existing resources be mobilized and managed for a more sustainable value creation?

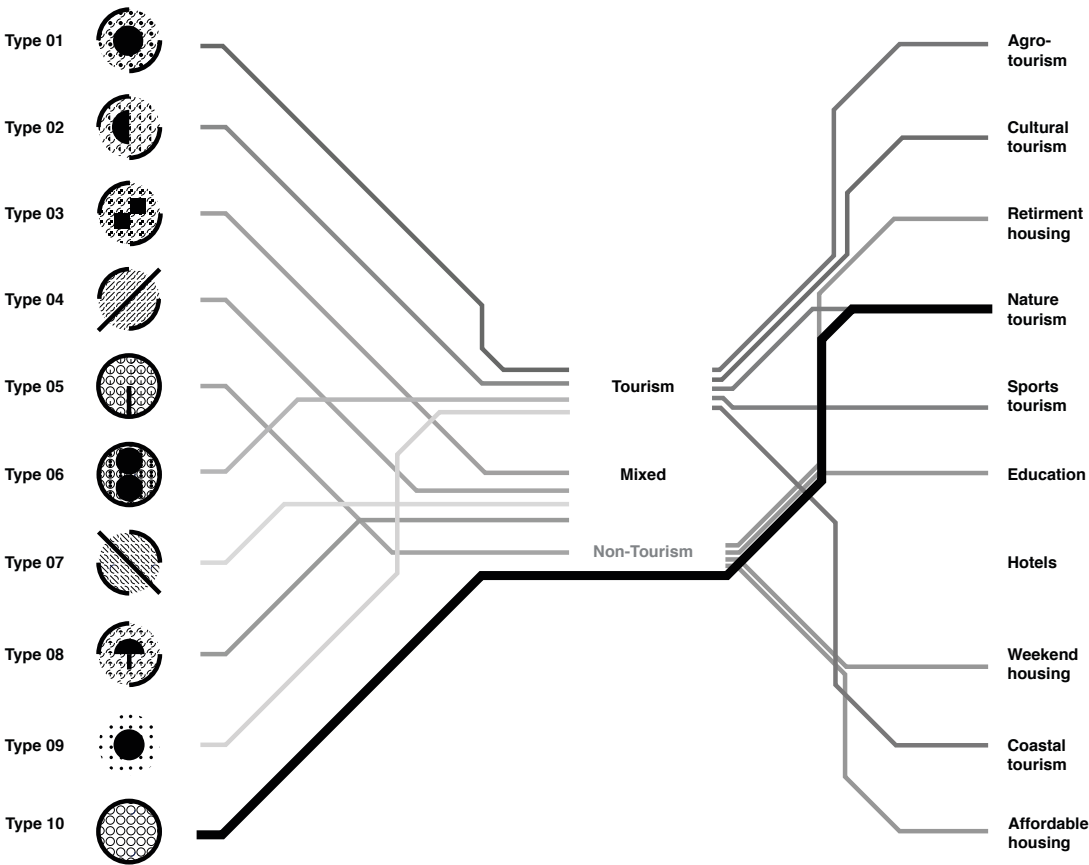
How would decentralisation strategies and renewal of the rural settlements inside the UNESCO region impact on its future?

Architecture case study;

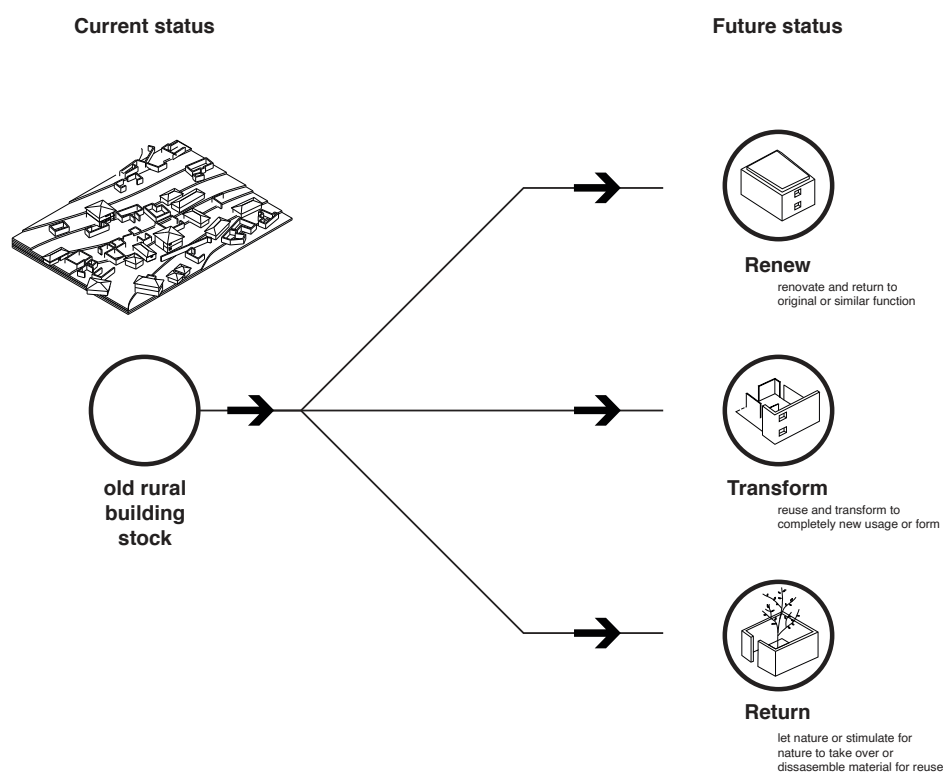
For a more tangible perspective and exploration with architectural discipline means, one settlement was chosen as a case study site, an abandoned village, that slowly starts to get non-formally built, as observed during the field trips. The proposal suggests Renew, Transform and Return framework for development. Where specific structures could be **Transformed**; to a new function, for example from what it used to be housing to a new function such as visitors centre, or its materials to be used for another function, **Renewed**; when a certain structure had the function for housing, it can stay to the same intended use, and **Returned**; when the structure is better off to integrate and go back to nature, this example might work the best with older houses that are mostly made of stone, wood, reeds and that do not pose any threat to its environment.

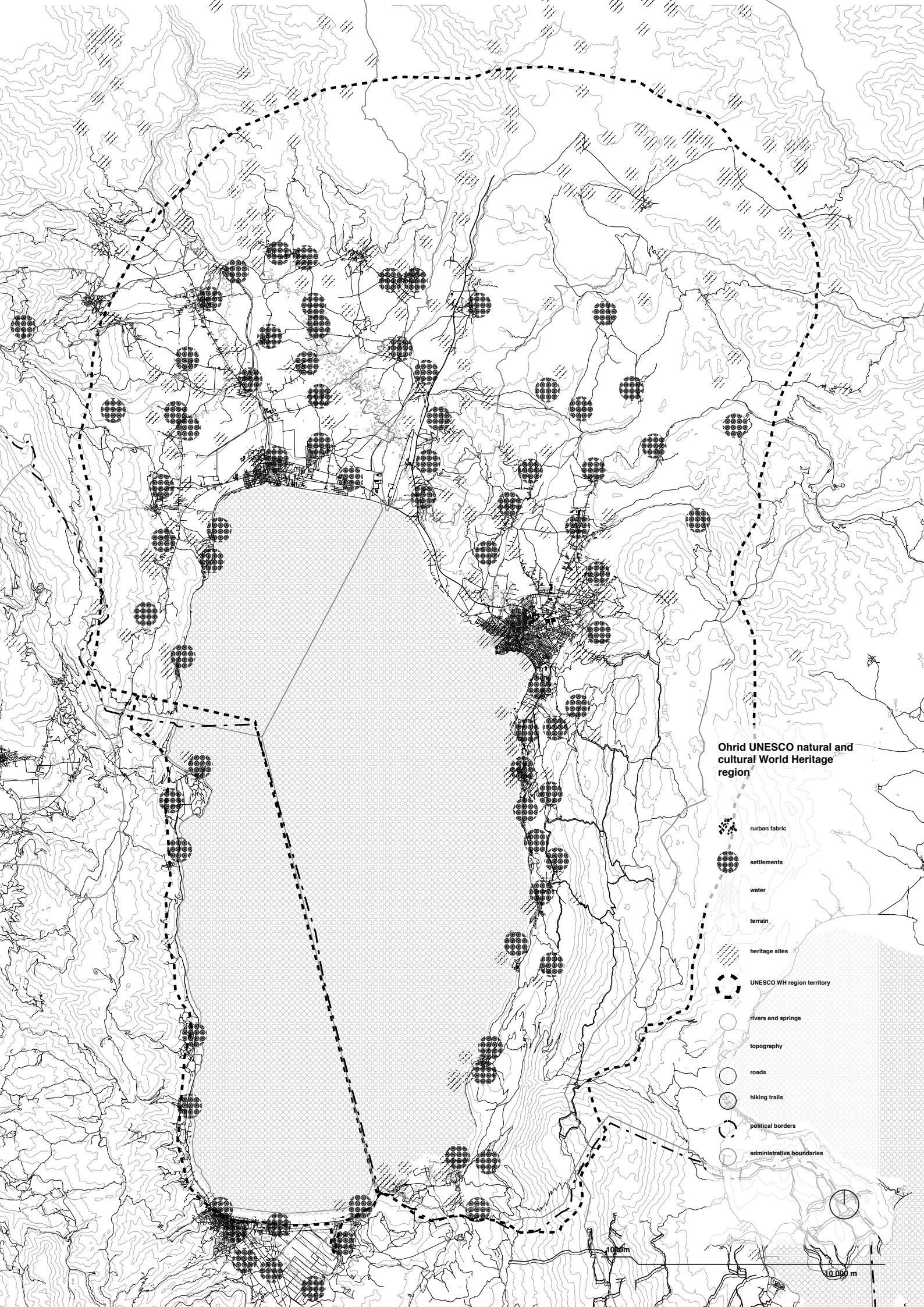
Additionally, the buildings assessments could be done by a conservation specialists, and use tools such as InterSAVE.(The Danish Nature Agency, 1997)





Different settlement typologies corresponding to specific future developmental type scenario, by autor, 2026





Typologies of rural settlements in the Ohrid UNESCO region

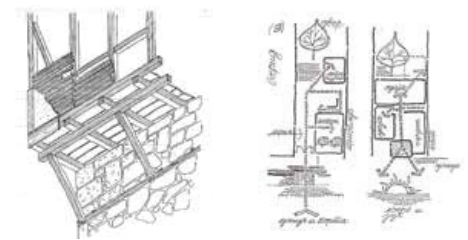
1:20 000
territorial heritage planning strategy



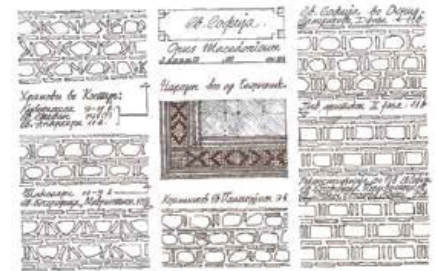
The choice of this settlement as a case study was informed both empirically, as well as bodily. This rural housing conglomeration lies on the east shore and overlooks the lake. It is 200 meters above the lake level, and 12,4 km from the main city, and it encapsulates the vernacular urban organic ensemble as well as traditional building techniques which overlap and streghned the UNESCO OUV criterion 1.

While being abandoned for the last 30+ years, durring the fieldwork constructing activites were observed. The proposal tries to integrate he heritage values, while also transforming the settlement to todays industry standards and incentives.

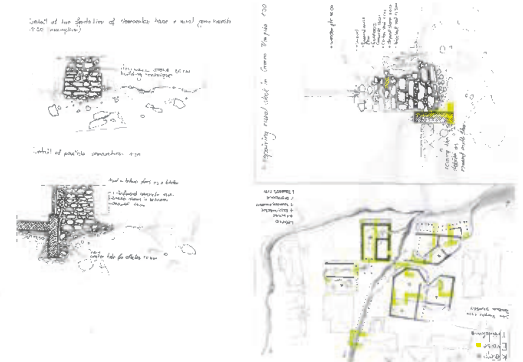
As it is part of the National Park, observed during the cartographic analisys, as well as wtih the initiatives of non-formal consturcting activities, and proximity to the lake, the proposal explores this settlments potential. Firstly, it suggests that the organic urban core it is kept, as it has human scale spatial configuration, distance between structure, narrow streets, and inbetween spaces, integration of the springs and watar canal into the urban setting. Secondly, on an architectural scale it suggests that the buidlings condition is evaluated, proceeding to their interveining to Return, Transform and Renew framework, depending on their conditions, as previously mentioned.



vernacular biling technologies that contribute to the heritage value, Cipan, 1995



vernacular biling technologies that contribute to the heritage value, Cipan, 1995



architecture iteration sketches for the case settlement, by author 2026



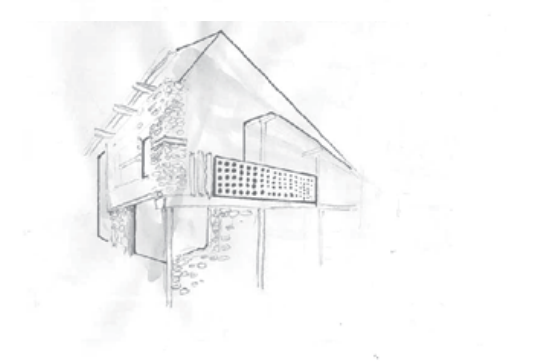
fieldtrip photos from the case study settlement, by author 2026



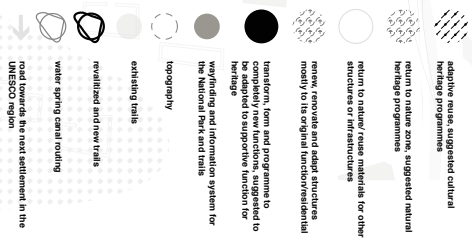
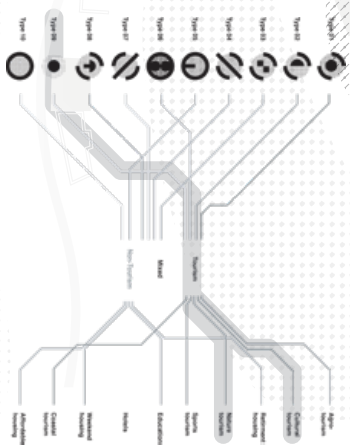
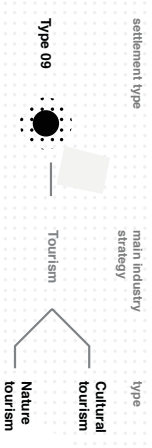
fieldtrip photos from the case study settlement, by author 2026

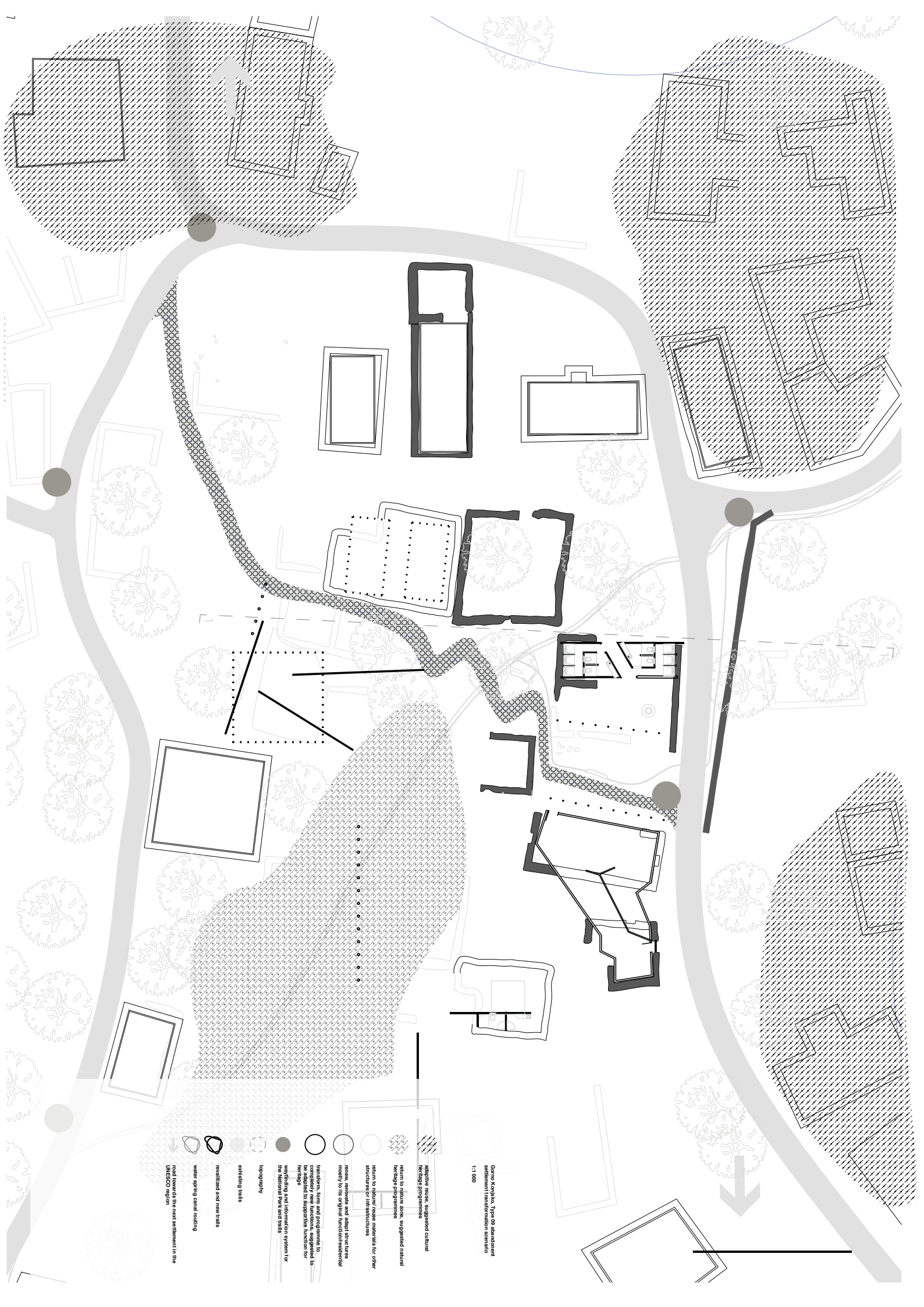


fieldtrip photos from the case study settlement, by author 2026



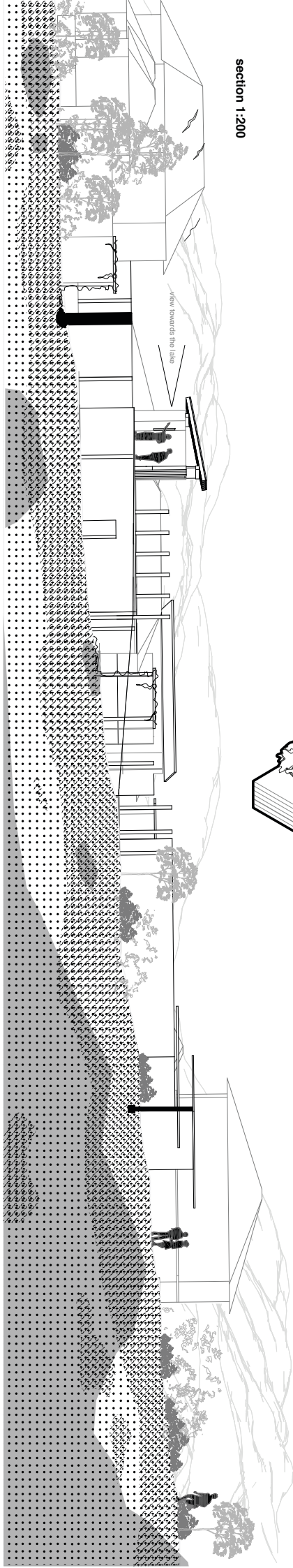
iterative architectural explorations, by author 2026



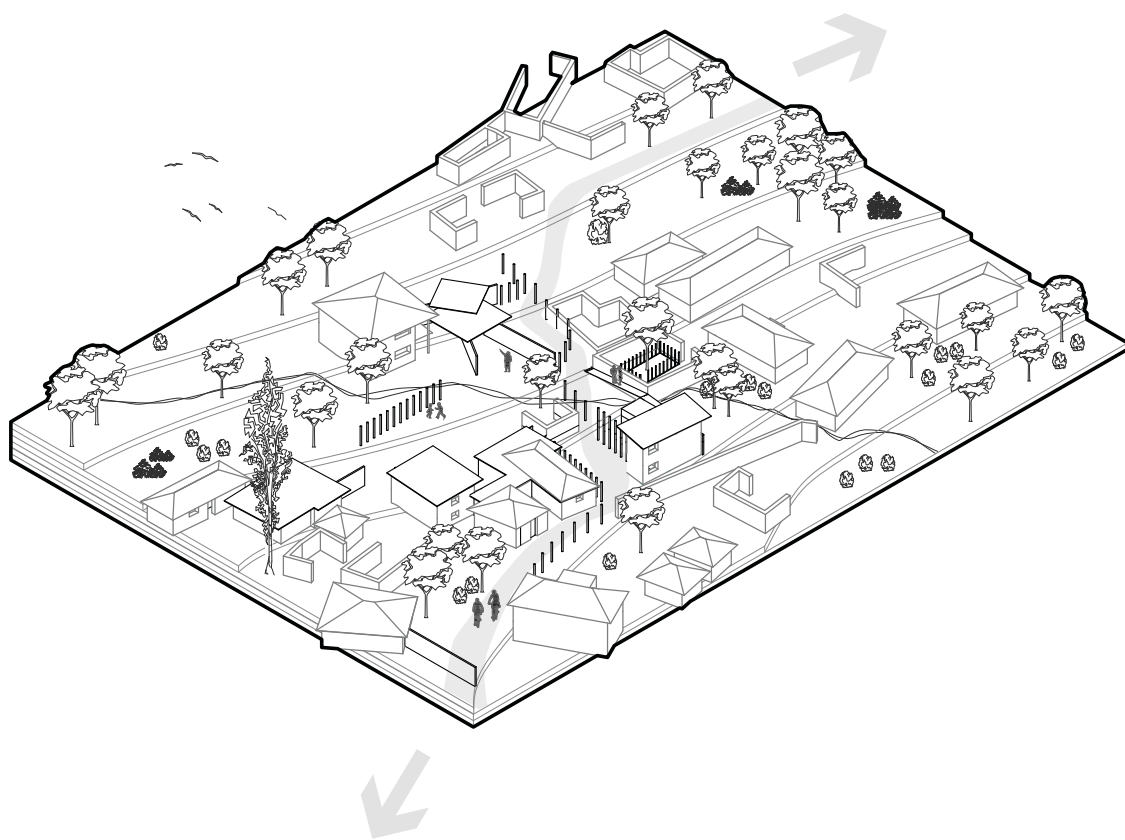


Gorno Kopsko, Type of abandoned
settlement transformation scenario
1:1 000

- adaptive reuse, suggested cultural
heritage programmes
- return to nature zone, suggested natural
heritage programmes
- return to nature, reuse materials for other
structures or infrastructures
- renew, renovate and adapt structures
mostly to its original function/residential
- transfer, form and programme to
completely new function, suggested to
heritage
- unifying and information system for
the national park and trails
- topography
- existing trails
- revalued and new trails
- water spring canal routing
- road layout of the next settlement in the
Jutišica region



section 1:200





visualisation of possible ways for debrec reuse and renewal, by author 2026



visualisation of possible ways for debrec reuse and hiking path by author 2026



visualisation of transformation of rural houses, by author 2026

CONCLUSIONS

While new apartment-housing typologies keep rising in the city and along the shoreline, the settlements across the UNESCO World Heritage region wait to be reactivated. They hold a real spatial and heritage capacity to be reused, renewed and transformed the resource is already there, in the existing building stock, rather than in further expansion.

The investment appetite now directed at apartment housing is itself a sign that the developmental motive and a supportive financial climate exist in the region. Redirected, that same energy points towards a scenario of distributive, decentralised development one that supports heritage values and provides spatial capacity at once, across diverse settlements that each keep their own place-based atmosphere.

Much of that value lies in the organic urban morphology of these settlements: the human scale, the gardens and greenery, the quality of the in-between spaces, and the inherited rule of a view. These are qualities the generic market-led typology cannot reproduce, because they grew from how these places were lived in rather than how fast they could be built. Many of these houses are themselves vernacular, forming part of the region's OUV under criterion (i).

A less acknowledged value is the local building technique, know-how and use of local materials far more sustainable than imported, market-led materials alien to the environment, and a basis for circularity within the region rather than a one-way import of resources.

Taken together, the World Heritage boundary is far larger, and holds far more potential, than is currently used. Reusing and transforming the existing stock with the right usage and strategy, and while keeping each settlement's original spatial properties does not only relieve pressure on the landscape; it can actively reinforce the Outstanding Universal Value under which the region was inscribed.

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